

Important Point: Why the Potential Definition Requires Path Independence

Potential is defined using a line integral

We defined the electrostatic potential (with reference at infinity) as

$$\phi(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}'$$

This formula uses a **line integral**, so we must address a key question:

Crucial requirement

For $\phi(\mathbf{r})$ to be a *well-defined single number* at the point $\mathbf{r}$, the value of the integral must be the *same* for **any path** chosen from ∞ to $\mathbf{r}$.

Intuitive explanation (why path dependence would be unacceptable)

Suppose you pick two different paths, C_1 and C_2 , from ∞ to the *same* point $\mathbf{r}$. If the line integral depended on the path, then you would get two different values:

$$\phi_{C_1}(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} \Big|_{C_1}, \quad \phi_{C_2}(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} \Big|_{C_2},$$

with $\phi_{C_1}(\mathbf{r}) \neq \phi_{C_2}(\mathbf{r})$. That would mean:

- The “potential at the point $\mathbf{r}$ ” is not unique.
- A scalar field $\phi(\mathbf{r})$ cannot be assigned consistently.
- The potential difference between two points would become ambiguous (depends on the chosen route).

Closed-loop viewpoint (a clean test)

Take the two paths C_1 and C_2 between the same endpoints, and traverse C_1 forward and C_2 backward. This forms a closed loop C :

$$\int_{C_1} \mathbf{E} \cdot d\mathbf{l} - \int_{C_2} \mathbf{E} \cdot d\mathbf{l} = \oint_C \mathbf{E} \cdot d\mathbf{l}.$$

Therefore:

$$\text{Path independence between endpoints} \iff \oint_C \mathbf{E} \cdot d\mathbf{l} = 0 \text{ for every closed loop } C.$$



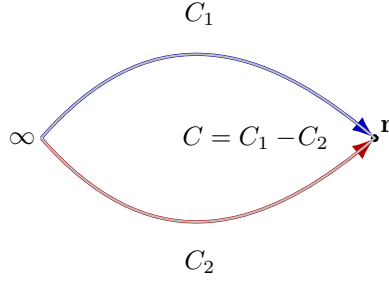


Figure 1: If two different paths C_1 and C_2 to the same point $\mathbf{r}$ gave different values, the potential at $\mathbf{r}$ would not be unique. Path-independence is equivalent to zero closed-loop integral $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.

Electrostatics statement (used later)

In electrostatics (time-independent fields), the electric field is conservative, so

$$\boxed{\oint \mathbf{E} \cdot d\mathbf{l} = 0} \quad \Rightarrow \quad \boxed{\phi(\mathbf{r}) \text{ is well-defined and single-valued.}}$$

(We will later connect this to $\nabla \times \mathbf{E} = \mathbf{0}$ and $\mathbf{E} = -\nabla\phi$.)

Key Takeaways

- Potential is defined by a line integral: $\phi(\mathbf{r}) = -\int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$.
- For $\phi(\mathbf{r})$ to be a unique scalar at a point, the integral must be path independent.
- Path independence between endpoints is equivalent to $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for any closed loop.

Advantage of the Potential Formulation

1) Stokes' theorem and the electrostatic condition

For any sufficiently smooth vector field $\mathbf{E}$ and any surface S bounded by a closed curve $C = \partial S$, Stokes' theorem states

$$\boxed{\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot \hat{\mathbf{n}} dS.}$$

In electrostatics we have already argued that the line integral over every closed loop is zero:

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0 \quad \text{for all closed } C.$$

Therefore, by Stokes' theorem, the surface integral must vanish for any surface S , which implies

$$\boxed{\nabla \times \mathbf{E} = \mathbf{0}.}$$



A field with zero curl is called a **conservative field**. Hence, electrostatic fields are a *special class* of vector fields.

2) Not every vector field can be an electrostatic field (example)

Consider the vector field

$$\mathbf{E} = y \hat{\mathbf{x}} \quad (\text{equivalently } \mathbf{E} = (y, 0, 0)).$$

Compute its curl:

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & 0 & 0 \end{vmatrix} = \left(\frac{\partial 0}{\partial y} - \frac{\partial 0}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial y}{\partial z} - \frac{\partial 0}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial 0}{\partial x} - \frac{\partial y}{\partial y} \right) \hat{\mathbf{z}}.$$

Since $\frac{\partial y}{\partial y} = 1$, we get

$$\nabla \times \mathbf{E} = -\hat{\mathbf{z}} \neq \mathbf{0}.$$

Hence this $\mathbf{E}$ **cannot** be an electrostatic electric field.

3) Why the scalar potential is so powerful

If $\nabla \times \mathbf{E} = \mathbf{0}$ in a simply connected region, then there exists a scalar potential $\phi(\mathbf{r})$ such that

$$\mathbf{E} = -\nabla\phi.$$

This is the key simplification:

Vector $\rightarrow$ scalar reduction

Instead of solving for three components (E_x, E_y, E_z) , we solve for *one* scalar function ϕ , then obtain the field by differentiation: $\mathbf{E} = -\nabla\phi$.

4) Component-level consequences of $\nabla \times \mathbf{E} = \mathbf{0}$

Writing $\nabla \times \mathbf{E} = \mathbf{0}$ in Cartesian components gives three equations:

$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0, \quad \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = 0, \quad \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = 0.$$

Equivalently (rearranging),

$$\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}, \quad \frac{\partial E_y}{\partial z} = \frac{\partial E_z}{\partial y}, \quad \frac{\partial E_z}{\partial x} = \frac{\partial E_x}{\partial z}.$$

These are **compatibility conditions**: an arbitrary vector field generally does *not* satisfy them, but an electrostatic field must.

Key Takeaways

- Stokes: $\oint_C \mathbf{E} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{E}) \cdot \hat{\mathbf{n}} dS$.
- Electrostatics: $\oint_C \mathbf{E} \cdot d\mathbf{l} = 0 \Rightarrow \nabla \times \mathbf{E} = \mathbf{0}$ (conservative field).
- Not every vector field is electrostatic (e.g., $\mathbf{E} = y\hat{\mathbf{x}}$ has $\nabla \times \mathbf{E} = -\hat{\mathbf{z}} \neq 0$).
- The potential formulation reduces a **vector** problem to a **scalar** problem: solve for ϕ , then $\mathbf{E} = -\nabla\phi$.

Reference Point for Potential: Only Potential Differences Matter

1) Potential depends on the chosen reference

We defined the electrostatic potential at a point $\mathbf{r}$ using a reference point $\mathbf{r}_{\text{ref}}$:

$$\phi(\mathbf{r}) = - \int_{\mathbf{r}_{\text{ref}}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}.$$

Often we choose $\mathbf{r}_{\text{ref}} = \infty$ and set $\phi(\infty) = 0$, but this is a *choice*, not a necessity.

2) Changing the reference shifts the potential by a constant

Suppose we choose a new reference point $\mathbf{r}'_{\text{ref}}$ and define

$$\phi'(\mathbf{r}) = - \int_{\mathbf{r}'_{\text{ref}}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}.$$

Insert an intermediate point $\mathbf{r}_{\text{ref}}$ (the old reference) in the path of integration:

$$\phi'(\mathbf{r}) = - \left[\int_{\mathbf{r}'_{\text{ref}}}^{\mathbf{r}_{\text{ref}}} \mathbf{E} \cdot d\mathbf{l} + \int_{\mathbf{r}_{\text{ref}}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} \right].$$

Define the constant

$$K \equiv - \int_{\mathbf{r}'_{\text{ref}}}^{\mathbf{r}_{\text{ref}}} \mathbf{E} \cdot d\mathbf{l},$$

which depends only on the two reference points (not on $\mathbf{r}$). Then

$$\phi'(\mathbf{r}) = K + \phi(\mathbf{r}).$$

So, changing the reference point simply **adds a constant** to the potential everywhere.



3) Potential differences are invariant

Take two points $\mathbf{r}_1$ and $\mathbf{r}_2$. Then

$$\phi'(\mathbf{r}_2) - \phi'(\mathbf{r}_1) = (K + \phi(\mathbf{r}_2)) - (K + \phi(\mathbf{r}_1)) = \phi(\mathbf{r}_2) - \phi(\mathbf{r}_1).$$

Hence,

$$\Delta\phi_{12} \text{ does not depend on the chosen reference.}$$

This is why **potential differences** (voltages) are the physically meaningful quantities.

4) Physical interpretation (altitude analogy)

The absolute value of potential is like **altitude**. We often measure altitude relative to mean sea level, but we could choose any other reference level. Changing the reference shifts all altitude values by a constant, but the *height difference* between two points remains the same.

Key Takeaways

- Potential is defined relative to a reference: $\phi(\mathbf{r}) = - \int_{\mathbf{r}_{\text{ref}}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$.
- Changing the reference adds a constant: $\phi'(\mathbf{r}) = \phi(\mathbf{r}) + K$.
- Potential differences are invariant: $\phi'(\mathbf{r}_2) - \phi'(\mathbf{r}_1) = \phi(\mathbf{r}_2) - \phi(\mathbf{r}_1)$.
- Physics uses voltages (differences), not absolute potential values.

Choosing the Reference at Infinity: A Special Case Where It Fails

1) Usual convention (works for localized charge distributions)

By analogy with altitude (measured relative to sea level), we often choose a convenient reference for potential. In electrostatics, the most common choice is

$$\phi(\infty) = 0,$$

which corresponds to taking the “far away” point as the zero-potential reference. This works well for *localized* charge distributions (finite total charge confined to a finite region), because the electric field typically decays to zero as $r \rightarrow \infty$ and the integral

$$\phi(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$$

converges.

2) Special case: an (ideal) infinite plane sheet of charge

Consider an infinite sheet with uniform surface charge density σ on the plane $z = 0$. The electrostatic field magnitude is constant:

$$\mathbf{E} = \begin{cases} +\frac{\sigma}{2\varepsilon_0} \hat{\mathbf{z}}, & z > 0, \\ -\frac{\sigma}{2\varepsilon_0} \hat{\mathbf{z}}, & z < 0. \end{cases}$$

Notice that $\|\mathbf{E}\| = \sigma/(2\varepsilon_0)$ does *not* go to zero as $|z| \rightarrow \infty$.

3) Consequence: potential relative to infinity diverges

Take a point on the $+z$ -axis. Then

$$\phi(z) - \phi(z_0) = - \int_{z_0}^z \mathbf{E} \cdot d\mathbf{l} = - \int_{z_0}^z \left(\frac{\sigma}{2\varepsilon_0} \hat{\mathbf{z}} \right) \cdot (\hat{\mathbf{z}} dz) = -\frac{\sigma}{2\varepsilon_0} \int_{z_0}^z dz.$$

Hence,

$$\phi(z) - \phi(z_0) = -\frac{\sigma}{2\varepsilon_0} (z - z_0).$$

If we try to take $z_0 \rightarrow \infty$ and impose $\phi(\infty) = 0$, the integral does not converge because $\mathbf{E}$ is constant:

$$\phi(z) = - \int_{\infty}^z \frac{\sigma}{2\varepsilon_0} dz \quad \text{diverges.}$$

So, for an infinite sheet, choosing “infinity” as the reference is **not convenient** (and not meaningful in the usual sense).

4) What we do instead

For such idealized infinite structures, we choose a **finite reference point**, e.g.,

$$\phi(0) = 0 \quad \text{or} \quad \phi(z_0) = 0 \text{ for some convenient } z_0,$$

and then use

$$\phi(z) = \phi(z_0) - \frac{\sigma}{2\varepsilon_0} (z - z_0).$$

Importantly, **potential differences** remain well-defined and physically meaningful even when the absolute potential relative to infinity is not.

5) Practical remark

In real physical systems there is no truly infinite sheet of charge; any actual charged surface is finite. Therefore, the $\phi(\infty) = 0$ convention generally works for realistic problems, and the “infinite plane” case should be understood as an idealization that is valid only over a limited region.

Key Takeaways

- For localized charge distributions, it is convenient to set $\phi(\infty) = 0$.
- For an ideal infinite sheet, $\|\mathbf{E}\| = \sigma/(2\epsilon_0)$ is constant and does not vanish at infinity.
- Therefore $\phi(z) = -\int_{\infty}^z \mathbf{E} \cdot d\mathbf{l}$ diverges; infinity is not a good reference.
- Choose a finite reference (e.g., $\phi(z_0) = 0$); potential differences remain meaningful.

Path Independence When the Source Charge Is Not at the Origin

Geometry and notation (observer vs source)

Let the observation point be $\mathbf{r}$ (“observer”) and let a point charge Q be located at the fixed source position $\mathbf{r}_Q$ (“source”). Define the displacement from the charge to the observer:

$$\mathbf{R} \equiv \mathbf{r} - \mathbf{r}_Q, \quad R = \|\mathbf{R}\|.$$

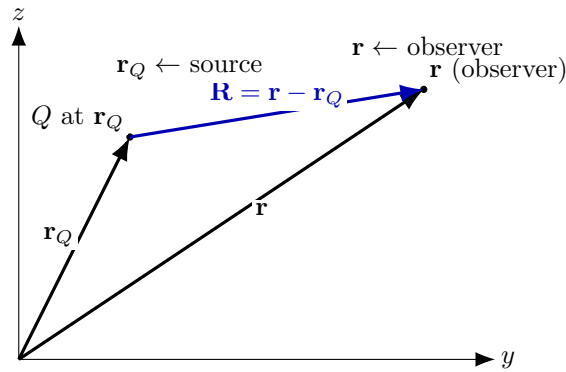


Figure 2: Shifted source: the field depends on $\mathbf{R} = \mathbf{r} - \mathbf{r}_Q$ (vector from source to observer).

Electric field of a shifted point charge

The Coulomb field at $\mathbf{r}$ due to a charge Q located at $\mathbf{r}_Q$ is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} Q \frac{\mathbf{r} - \mathbf{r}_Q}{\|\mathbf{r} - \mathbf{r}_Q\|^3} = \frac{1}{4\pi\epsilon_0} Q \frac{\mathbf{R}}{R^3}.$$

Path independence via a coordinate translation (move the origin to the charge)

Consider moving a test charge q from A to B along any path C . The work done by the electric force is

$$W_C(A \rightarrow B) = \int_C \mathbf{F} \cdot d\mathbf{l} = q \int_C \mathbf{E}(\mathbf{r}) \cdot d\mathbf{l}.$$



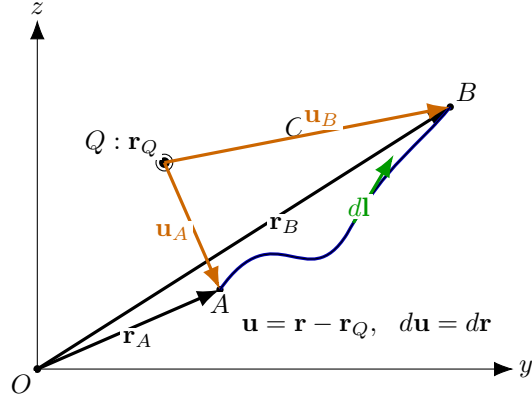


Figure 3: Coordinate transformation used in the proof: Black vectors are $\mathbf{r}_A, \mathbf{r}_B$ measured from the original origin O . Orange vectors are $\mathbf{u}_A = \mathbf{r}_A - \mathbf{r}_Q$ and $\mathbf{u}_B = \mathbf{r}_B - \mathbf{r}_Q$ measured from the charge location Q . Along the same physical path C , $d\mathbf{u} = d\mathbf{r} = d\mathbf{l}$ since $\mathbf{r}_Q$ is constant.

Now perform the **translation**

$$\mathbf{u} \equiv \mathbf{r} - \mathbf{r}_Q \Rightarrow d\mathbf{u} = d\mathbf{r} = d\mathbf{l} \quad (\mathbf{r}_Q \text{ is constant}).$$

Then $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} Q \frac{\mathbf{u}}{\|\mathbf{u}\|^3}$ and

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \frac{Q}{4\pi\epsilon_0} \int_C \frac{\mathbf{u}}{\|\mathbf{u}\|^3} \cdot d\mathbf{u}.$$

Use the identity

$$d\|\mathbf{u}\| = \frac{\mathbf{u} \cdot d\mathbf{u}}{\|\mathbf{u}\|} \Rightarrow \frac{\mathbf{u} \cdot d\mathbf{u}}{\|\mathbf{u}\|^3} = \frac{d\|\mathbf{u}\|}{\|\mathbf{u}\|^2}.$$

Therefore,

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \frac{Q}{4\pi\epsilon_0} \int_{u_A}^{u_B} \frac{1}{u^2} du = \frac{Q}{4\pi\epsilon_0} \left[-\frac{1}{u} \right]_{u_A}^{u_B},$$

where $u_A = \|\mathbf{r}_A - \mathbf{r}_Q\|$ and $u_B = \|\mathbf{r}_B - \mathbf{r}_Q\|$. Hence,

$$\boxed{\int_C \mathbf{E} \cdot d\mathbf{l} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{\|\mathbf{r}_A - \mathbf{r}_Q\|} - \frac{1}{\|\mathbf{r}_B - \mathbf{r}_Q\|} \right)}$$

and this depends **only** on the endpoints (not on the path). Consequently, electrostatic potential is well-defined even when the charge is not at the origin.

Potential of a shifted point charge

Define ϕ such that $\phi_B - \phi_A = -\int_A^B \mathbf{E} \cdot d\mathbf{l}$. Using the result above:

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\|\mathbf{r} - \mathbf{r}_Q\|} + \text{constant}.$$



With the common reference $\phi(\infty) = 0$ (for localized charges),

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\|\mathbf{r} - \mathbf{r}_Q\|}.$$

Superposition Using the Potential Viewpoint

1) Discrete charges

If there are N point charges Q_i located at $\mathbf{r}'_i$, the electric field is the sum:

$$\mathbf{E}(\mathbf{r}) = \sum_{i=1}^N \mathbf{E}_i(\mathbf{r}), \quad \mathbf{E}_i(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} Q_i \frac{\mathbf{r} - \mathbf{r}'_i}{\|\mathbf{r} - \mathbf{r}'_i\|^3}.$$

For any path C ,

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \int_C \left(\sum_{i=1}^N \mathbf{E}_i \right) \cdot d\mathbf{l} = \sum_{i=1}^N \int_C \mathbf{E}_i \cdot d\mathbf{l}.$$

Each $\int_C \mathbf{E}_i \cdot d\mathbf{l}$ is path independent (by the shifted-charge result above), so their sum is also path independent. Hence a single scalar potential exists, and

$$\phi(\mathbf{r}) = \sum_{i=1}^N \phi_i(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i}{\|\mathbf{r} - \mathbf{r}'_i\|}.$$

This is the cleanest way to see why **superposition works naturally** for potential.

2) Continuous volume charge density

Replace the discrete charges by a continuous distribution $\rho(\mathbf{r}')$ over volume V . A small volume element $d\tau'$ contains charge $dq = \rho(\mathbf{r}') d\tau'$ and contributes

$$d\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{dq}{\|\mathbf{r} - \mathbf{r}'\|} = \frac{1}{4\pi\epsilon_0} \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} d\tau'.$$

Integrating over all source points $\mathbf{r}'$:

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint_V \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} d\tau'.$$

Key Takeaways

- For a point charge at $\mathbf{r}_Q$, the field depends on $\mathbf{R} = \mathbf{r} - \mathbf{r}_Q$.

- A translation $\mathbf{u} = \mathbf{r} - \mathbf{r}_Q$ shows

$$\int_C \mathbf{E} \cdot d\mathbf{l} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{\|\mathbf{r}_A - \mathbf{r}_Q\|} - \frac{1}{\|\mathbf{r}_B - \mathbf{r}_Q\|} \right),$$

which is path independent.

- Therefore $\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{\|\mathbf{r} - \mathbf{r}_Q\|}$ (with $\phi(\infty) = 0$).
- Superposition follows immediately: potentials add for multiple charges.
- Continuum limit gives $\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint \frac{\rho(\mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} d\tau'$.